NAVYA BOGA

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Skills

- **Languages**: Python, Java, SQL, C++
- **Backend & APIs:** Node.js, REST APIs, Microservices, GraphQL, API Integration, Workflow Automation
- **Front-end**: React.js, TypeScript, JavaScript, Redux Toolkit (RTK Query), HTML, CSS, AWS Cloudscape Design System
- **Cloud & Infrastructure**: AWS, AWS CDK, CloudFormation, Dockier, Kubernetes, Linux, CI/CD, Distributed Systems, CloudWatch, Infrastructure Monitoring, Access Controls, S3, EC2
- **AI/ML & GenAI**: Generative AI (GenAI), Scalable Model Inference, TensorFlow, OpenCV, Transfer Learning, Image Processing, Anomaly Detection, Computer Vision, Rapid Prototyping
- **Testing & Tools**: Vitest, React Testing Library, Mockito, Jenkins, Git, Agile Development, AI-Assisted Development (Claude Code)

Education

- Master of Science - Data Science **The University of Memphis** Memphis, TN, USA (08/2023 - 05/2025)
- Bachelor of Engineering – Computer Science **Osmania University** (05/2019 - 07/2023)

Experience

Software Development Engineer - Amazon Web Services, S3 Console **Jul 2025 – Present**

• Built and launched full-stack platform features for the AWS S3 Console used by millions of customers, delivering tooling and APIs that other teams customize and build on, across React/TypeScript frontend, backend integrations, and cloud infrastructure.
• Architected reusable Storage Lens workflows with Redux Toolkit state management, validation, and API-driven resource orchestration.
• Prototyped and iterated on reusable component architectures and API abstractions based on user feedback, standardizing workflows and packaging them for reuse across multiple teams and features.
• Developed multi-region resource management workflows with AWS Organizations integration, org-scoped access controls, and concurrent cross-region data fetching with resilient partial-failure handling.
• Engineered analytics dashboards and anomaly detection across 19 storage metrics, enabling data-driven insights and monitoring.
• Improved deployment reliability through AWS CDK deployment-wave isolation for sovereign cloud regions and expanded infrastructure monitoring and canary coverage across Gamma and Production environments.
• Resolved production issues involving state synchronization, deployment pipelines, and infrastructure reliability through debugging and system improvements.
• Developed integration and infrastructure test suites using Vitest, React Testing Library, and Mockito, maintaining 99%+ test coverage across new feature development.
• Delivered infrastructure monitoring and observability tooling with CloudWatch dashboards and reporting across distributed, multi-region compute environments.
Technologies: React.js, TypeScript, Redux Toolkit, CloudFormation, CloudWatch, S3, EC2, Java, Node.js, Vitest, Mockito

Graduate Assistant - Center for Information Assurance, The University of Memphis **Dec 2024 – May 2025**

• Developed and maintained web-based cybersecurity training modules and instructional platforms using Python, JavaScript, HTML/CSS, and data visualization tools, supporting coursework focused on network defense, risk management, and incident response.
• Built interactive dashboards and technical content delivery workflows for cybersecurity research and training under the National Cybersecurity Preparedness Consortium (NCPC) in collaboration with the Department of Homeland Security.

Data Engineer Intern - The University of Tennessee Health Science Center **May 2024 - Aug 2024**

• Performed large-scale genomic data analysis and preprocessing using Python, R, pandas, and NumPy, improving mutation detection accuracy across mitochondrial DNA datasets. Optimized data processing and visualization workflows using Jupyter Notebook, Matplotlib, Shell scripting, and Unix-based tooling. **Technologies:** Python, R, Pandas, NumPy, Unix, Shell Scripting

Java Ignite Trainee and Intern Larsen & Toubro Infotech Mindtree **Nov 2022 - Jul 2023**

• Developed Java-based cloud applications using AWS services including EC2, S3, RDS, and Lambda for scalable deployment workflows and backend integrations. Built and maintained CI/CD pipelines using Jenkins, Docker, Kubernetes, and Git, reducing deployment errors, improving operational efficiency.
• Worked on REST API integrations, Agile development workflows, and monitoring automation for distributed application environments.
Technologies: Java, AWS, Docker, Kubernetes, Jenkins, REST APIs, Git

Web Application Developer Intern - Suntek Corporations Private Limited **Jul 2022 - Aug 2022**

• Built a facial recognition attendance system using Python, OpenCV, NumPy, and machine learning techniques including CNN, SVM, and HOG, improving image processing performance and reducing false identification rates. (REPOSITORY)

Web Application Developer Intern - Defense Research and Development Organization **May 2022 – Jun 2022**

• Developed a real-time intranet chat application using React.js, PHP, SQL, Python, WebSockets, and Django Channels, supporting 100+ concurrent users with low-latency secure messaging. (REPOSITORY)

Projects

• **Forged Emblem Detection System:** Designed to identify counterfeit emblems with 97.5% accuracy using advanced image processing techniques (Transfer Learning) to ensure accurate detection and authentication of legitimate emblems. **Tech Stack**: Python, TensorFlow, PIL (Python Imaging Library), Streamlit, NumPy, SSL, network security and Infrastructure system design. (REPOSITORY)
• **E-Commerce Platform Development**: Engineered a full-fledged e-commerce system with secure user authentication, dynamic product listings, and seamless checkout and payment integration. Achieved optimized data handling and performance using MySQL and Neo4j for scalability. **Tech Stack:** React.js, Node.js, MySQL, Neo4j, Stripe API, PHP, SSL encryption, and RESTful API design. (REPOSITORY)

Courses and Certifications

- **ORACLE**: Cloud Infrastructure 2024 Generative AI Certified Professional
- **AWS**: Cloud Foundations